Evolution of communication in populations of honeybee-like agents

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1 Introduction

Honeybee waggle dances provide spatial information about resources away from the nest. One ecological hypothesis is that such communication is favored when food is difficult to discover independently but remains useful after a worker has found it. Empirical and theoretical studies suggest that the value of dance information and recruitment depends on the spatiotemporal distribution of food resources, including patch density, patch size, reward variability, and habitat [7, 3, 4, 1, 2, 6, 5]. The model is deliberately minimal: it asks which evolutionary pressures are needed before adding richer social, developmental, or landscape detail. We start from direct directional communication on an exposed horizontal comb, where dance angle can point in the food direction, and then ask what additional conditions make a gravity-referenced vertical-comb code reachable.

The simulations evaluate four evolutionary hypotheses.

1. **Resource-distribution hypothesis.** Costly directional communication should be favored when independent discovery is difficult enough that recruitment is useful, but not so difficult that few dances are seeded. It should be weakly favored when food is very easy to find without dances.

2. **Architectural-precondition hypothesis.** Direct pointing should work well on a horizontal comb, whereas gravity-referenced communication should become useful only when the comb geometry supplies a reliable gravity cue. A non-communication benefit of vertical combs may therefore be needed to make the gravity channel selectively available.

3. **Sender–receiver coordination hypothesis.** Sender and receiver transposition must evolve together. Positive coupling between sender and receiver mutations should reduce the time spent in mismatched codes.

4. **Transition-risk hypothesis.** Selection for vertical combs can outpace adaptation of the communication code. Successful transitions should therefore require overlap among vertical geometry, directional precision, receiver attention, and matched sender–receiver transposition; otherwise foraging may collapse before the new code becomes useful.

2 Model

2.1 Conceptual model

Colonies are the reproducing entities, while workers are behavioral agents sampled from colony-level trait means. A colony has seven heritable mean traits: directional precision investment, receiver attention, sender transposition, receiver transposition, search limit, comb tilt, and comb orientation. Directional precision investment is a costly investment in producing directionally informative dances. Receiver attention controls whether workers use available dances. Sender and receiver transposition interpolate between two communication modes: direct pointing in the comb plane and sun/gravity-referenced encoding. Comb tilt and orientation define the dance plane in the world, and search limit determines how far a worker searches along its chosen direction.

Each foraging episode samples food sites with direction, distance, angular width, value, and capacity. Workers act sequentially. If dances are available, a worker may follow one; otherwise it searches independently in a random direction. Independent searchers that find food add a noisy dance for that site. There are therefore no fixed scouts in the model. Background discovery and recruitment both emerge from the food distribution, receiver attention, search limits, and dance precision. This follows ecological recruitment models in which communication is valuable only when resources are discoverable and worth recruiting to [4, 1].

Comb geometry determines which directional cue is available. Direct pointing projects the horizontal food direction onto the comb plane, so its strength depends on the food direction, comb tilt, and comb orientation. Gravity- referenced mapping uses the projection of gravity onto the comb plane together with the episode's sun azimuth. Gravity provides no in-plane reference on a horizontal comb and becomes a stronger cue as the comb becomes vertical. Comb orientation can be treated as circular, or as axial when orientations ϕ and $\phi + 180°$ represent the same comb plane.

2.2 Mathematical implementation

Let a colony's mean strategy be

$$\theta = (b, a, s, r, \ell, t, \phi),$$

where $b \in [0, 1]$ is directional precision investment, $a \in [0, 1]$ is receiver attention, $s, r \in [0, 1]$ are sender and receiver transposition, $\ell \geq 0$ is search limit, $t \in [0, 1]$ is comb tilt, and ϕ is comb orientation. A worker i receives stable within-colony deviations from the colony means:

$$\theta_i = \text{clip}(\theta + \epsilon_i).$$

These deviations represent worker heterogeneity but are not themselves heritable.

For a food direction α and sun azimuth ψ, let $D(\alpha; t, \phi)$ be the direct projected dance angle and $G(\alpha, \psi; t, \phi)$ be the sun/gravity-referenced dance angle. Let q_D and q_G be their corresponding cue strengths in the comb plane. A sender with transposition s_i encodes the dance mean angle as a weighted circular mean,

$$\mu_i = \text{cmean}\left((D, (1 - s_i)q_D),\ (G, s_i q_G)\right).$$

The produced dance signal is

$$x_i = \text{VM}(\mu_i, \kappa_{\max} b_i) + \eta_d \pmod{2\pi},$$

where VM is the von Mises distribution, $\kappa_{\max} = 14$ in the reported simulations, and η_d is transient dance-production noise. A receiver with transposition r_j applies the analogous weighted decoding rule and adds interpretation noise η_r to obtain a search direction $\hat{\alpha}_j$.

A worker succeeds at food site k only if

$$\Delta(\hat{\alpha}_j, \alpha_k) \leq w_k \quad \text{and} \quad d_k \leq \ell_j,$$

where Δ is circular angular distance, w_k is site angular width, and d_k is site distance. If multiple available sites satisfy this condition, the nearest site is used.

Let the net foraging payoff for an episode be

$$F_e = \sum_{j \in \mathcal{F}_e} v_{k(j)} - c_T \left(\sum_{j \in \mathcal{F}_e} d_{k(j)} + \sum_{j \in \mathcal{U}_e} \ell_j \right) - \sum_{i \in \mathcal{D}_e} (c_0 + c_b b_i) - c_a A_e.$$

Here $\mathcal{F}_e$ is the set of successful workers, $\mathcal{U}_e$ the unsuccessful workers, $\mathcal{D}_e$ the independent finders that produce dances, and A_e the number of dance-following attempts. The terms are food value, travel/search cost, per-dance communication cost, and receiver-attention cost. A vertical comb can then have an optional non-communication advantage that scales, rather than adds to, net performance:

$$P_e = F_e \left(1 + \alpha f(t)\right).$$

The default modifier is $f(t) = t$, so $\alpha = 0.05$ means that a fully vertical comb has a 5% advantage over a horizontal comb with the same foraging performance. The long-transition robustness check also uses $f(t) = \mathbf{1}[t \geq 0.8]$. Because the advantage is multiplicative, verticality cannot rescue a colony whose foraging payoff has collapsed. Mean colony payoff is averaged over episodes and floored at 0.001 so that low-payoff colonies remain selectable but rare.

Daughter colonies are sampled in proportion to payoff. Only colony means are inherited. Mutations are Gaussian and clipped to the allowed ranges; sender and receiver transposition mutations can have correlation ρ, which is the coupling parameter used in the vertical-transition probes.

2.3 Simulation parameters and reproducibility

Unless otherwise stated, reported runs use 60 colonies, 80 workers per colony, 50 foraging episodes per colony per generation, 12 foraging attempts per episode, ten random seeds, within-colony standard deviation 0.08 for unit-interval traits, mutation standard deviation 0.07, dance-production noise standard deviation 0.18, interpretation-noise standard deviation 0.12, maximum search distance 8.0, food value 1.0, food-site capacity 6, travel cost 0.02 per distance unit, precision-dependent dance-cost coefficient $c_b = 0.02$, baseline dance cost $c_0 = 0$, and attention cost $c_a = 0.01$. Search-limit noise and mutation use the same scales multiplied by the maximum search distance. Initial directional precision is sampled from $[0, 0.15]$, initial receiver attention from $[0, 0.25]$, initial transposition traits are 0, and initial search limit is sampled from 15–45% of the maximum search distance.

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```
python -u experiments/run_report_artifacts.py artifacts
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rebuilds the LaTeX fragments from the CSV files in `results/`; running the same script with `all` first regenerates the result CSVs. The provenance manifest is `report/artifact_sources.csv`.

3 Results

3.1 Horizontal comb

We first compared three simple food distributions over ten random seeds (101–110). The "hard" condition had one narrow food site per episode, the baseline condition had two moderately wide food sites, and the "easy" condition had many broad food sites. Table 1 reports the generation at which mean directional precision investment first reached 0.30, plus final-generation trait, success, and payoff values.

Each run used 60 colonies, 80 workers per colony, 30 generations, 50 foraging episodes per colony per generation, 12 foraging attempts per episode, and horizontal-comb tilt $t = 0$. Food-site distances were sampled uniformly from 1.0 to 8.0, maximum search distance was 8.0, travel cost was 0.02 per distance unit, baseline dance cost was 0, the precision-dependent dance-cost coefficient was 0.02, and food-site value was 1.0. All three conditions used food-site capacity 6. The conditions varied only the number and angular width of food sites: hard used one site of width 0.08, baseline used two sites of width 0.20, and easy used eight sites of width 0.50. The hard condition therefore makes independent discovery unlikely, while the easy condition makes independent discovery likely even without using the dance.

Condition	Reached	Reach gen.	Prec.	Attention	Search	Success	Payoff
Hard	4/10	22.750	0.264	0.313	2.789	0.008	0.001
Baseline	10/10	7.700	0.844	0.868	7.163	0.194	0.733
Easy	8/10	20.250	0.343	0.366	7.045	0.709	7.416

Table 1: Food-distribution experiment under the current dynamic recruitment model. “Reached” is the number of seeds in which mean directional precision investment reached 0.30 by generation 30. “Reach gen.” is averaged over reached seeds only. Prec. is final mean directional precision investment; attention, search limit, success, and payoff values are also final-generation means over all ten seeds.

The result supports the resource-distribution hypothesis. In the hard environment, food is so rare and narrow that independent discoveries seldom seed dances; final success and payoff remain close to zero, and communication crosses the directional-precision threshold in only 4 of 10 seeds. In the baseline environment, independent discovery is sufficient to seed recruitment cascades, and directional precision, receiver attention, and search limit all rise strongly. In the easy environment, success and payoff are high, but directional precision and receiver attention remain much lower than in the baseline case because food is often found without precise communication.

3.2 Vertical comb

We also compared colonies initialized with horizontal, tilted, and vertical combs while holding the baseline food distribution fixed. Initial comb tilt was set to $t = 0$, $t = 0.5$, or $t = 1.0$, with ten seeds per condition. Comb tilt was still heritable and mutable in these runs. Table 2 summarizes the result after 30 generations.

Comb	Reached	Reach gen.	Prec.	Attention	Sender	Receiver	Success	Payoff
Horizontal	10/10	7.700	0.844	0.868	0.187	0.176	0.194	0.733
Tilted	10/10	8.100	0.829	0.886	0.171	0.196	0.195	0.727
Vertical	10/10	16.500	0.606	0.794	0.384	0.362	0.165	0.414

Table 2: Comb-tilt comparison using the baseline food distribution and ten seeds. Sender and receiver denote final mean sender and receiver transposition traits.

With mutable tilt, all three initial-tilt conditions reach the directional-precision threshold in all ten seeds under the current baseline settings. Tilted and vertical starts retain substantial foraging success over 30 generations, but they also evolve higher sender and receiver transposition than horizontal starts. The initially vertical condition still has lower payoff than the horizontal condition, and its final mean tilt is reduced in the underlying result file, indicating that the short-run comparison does not by itself establish a stable vertical-comb transition.

3.3 Tilt-geometry calibration

We next ran a calibration grid over initial comb tilt and the proportional vertical-comb advantage α using the default modifier $f(t) = t$. These runs used the baseline food distribution, the daytime sun arc from 0° to 180° centered at 90°, and ten seeds (101–110). Success is the fraction of foraging attempts that found food. Payoff is the final mean colony payoff after food rewards, search/travel costs, dance costs, attention costs, and the multiplicative vertical-comb modifier. Table 3 shows that proportional advantages of 15–25% do not by themselves make horizontal or tilted starts retain verticality within 30 generations, but vertical starts can retain more tilt.

Initial tilt	α	Final tilt	Sender	Receiver	Success	Payoff	Reached
0.000	0.150	0.129	0.186	0.177	0.197	0.757	10/10
0.000	0.200	0.136	0.188	0.205	0.193	0.731	10/10
0.000	0.250	0.149	0.188	0.196	0.196	0.753	10/10
0.500	0.150	0.154	0.183	0.196	0.192	0.708	10/10
0.500	0.200	0.126	0.232	0.204	0.190	0.692	10/10
0.500	0.250	0.237	0.211	0.250	0.187	0.672	10/10
1.000	0.150	0.568	0.403	0.424	0.164	0.450	9/10
1.000	0.200	0.292	0.248	0.261	0.176	0.580	10/10
1.000	0.250	0.636	0.440	0.432	0.160	0.449	9/10

Table 3: Ten-seed calibration of the proportional vertical-comb advantage under the explicit tilt-geometry model. Sender and receiver are final mean transposition traits. Reached is the number of seeds in which mean directional precision investment reached 0.30 by generation 30.

The calibration suggests that the transition is strongly path-dependent. Starting from a horizontal or tilted comb, the tested advantages leave final mean tilt below 0.25 while preserving high foraging success. Starting from a vertical comb, final tilt is higher, especially at $\alpha = 0.15$ and $\alpha = 0.25$, but success and payoff are lower than in flatter conditions. The proportional rule therefore removes the artificial life-support effect of an additive bonus: verticality is retained only when enough foraging performance remains to be scaled. The relevant open question is whether longer runs can turn supplied verticality into a coordinated gravity-referenced code before selection drives colonies back toward flatter combs.

3.4 Tilt orientation

We also examined the evolved orientation of the comb tilt. The current sun arc is centered at 90°, so Table 4 reports the weighted mean orientation in degrees and its offset from this sun-arc center. The within-alignment column measures how strongly colonies within a run converge on an orientation. The across-alignment column measures whether different seeds converge on the same absolute orientation, weighting each seed by its within-run orientation alignment.

Orientation sometimes aligns within a run, but it does not show a clean, robust relationship to the daytime sun arc across seeds under the circular orientation representation. Some conditions point near the sun-arc center, some near the opposite direction, and several have weak across-seed alignment. This indicates weak or path-dependent selection on orientation in these probes and

Initial tilt	α	Final tilt	Success	Within align.	Mean orient.	Sun offset	Across align.
0.000	0.150	0.129	0.197	0.480	352.790	-97.2	0.182
0.000	0.200	0.136	0.193	0.396	174.322	84.3	0.092
0.000	0.250	0.149	0.196	0.461	263.759	173.8	0.431
0.500	0.150	0.154	0.192	0.430	95.069	5.1	0.289
0.500	0.200	0.126	0.190	0.506	337.558	-112.4	0.136
0.500	0.250	0.237	0.187	0.610	268.532	178.5	0.255
1.000	0.150	0.568	0.164	0.653	285.222	-164.8	0.490
1.000	0.200	0.292	0.176	0.591	305.511	-144.5	0.390
1.000	0.250	0.636	0.160	0.768	238.536	148.5	0.423

Table 4: Orientation sanity check under the explicit tilt-geometry model. Mean orientation and sun offset are measured in degrees. Across alignment is the weighted across-seed orientation alignment; values near 0 indicate that different seeds choose different absolute orientations.

motivates the axial representation used in the longer transition runs, where ϕ and $\phi + 180^\circ$ are treated as the same comb plane.

3.5 Constrained vertical coupling

The tilt-calibration results left open whether gravity-referenced communication is difficult because the sender–receiver code itself is hard to reach, or because colonies must evolve the vertical geometry and the sender–receiver code at the same time. To separate these possibilities, we ran a constrained probe with the geometric precondition supplied. Initial comb tilt was fixed near vertical at $t = 0.95$, comb-tilt mutation was set to 0, comb orientation was treated axially, the vertical-comb advantage was set to 0, and the baseline food distribution was otherwise retained. We then varied only the sender–receiver transposition mutation correlation and ran each condition for 120 generations over five seeds (101–105). A run was counted as reaching gravity-referenced communication when both mean sender and receiver transposition reached 0.50.

Corr.	Reached	Reach gen.	Prec.	Sender	Receiver	Gap	Success	Payoff
0.000	5/5	42.600	0.878	0.864	0.881	0.033	0.192	0.680
0.300	5/5	46.200	0.847	0.913	0.898	0.022	0.192	0.694
0.600	5/5	21.000	0.882	0.890	0.897	0.010	0.199	0.754
0.900	5/5	22.400	0.863	0.907	0.903	0.005	0.194	0.723
1.000	5/5	23.400	0.836	0.901	0.901	0.000	0.191	0.677

Table 5: Constrained near-vertical coupling probe. Corr. is the correlation between sender and receiver transposition mutation increments. "Reached" counts seeds in which both mean sender and receiver transposition reached 0.50 by generation 120. Reach generation is averaged over reached seeds only. Gap is the final mean absolute sender–receiver transposition difference.

Table 5 separates reachability from transition speed. With near-vertical geometry supplied, every correlation condition reaches gravity-referenced communication in all five seeds, so the gravity channel is not intrinsically unreachable. The main difference is how quickly the matched sender–receiver code appears. Independent or weakly coupled mutations ($\rho = 0$ or 0.3) reach the threshold only after about 43–46 generations on average. Moderate to strong coupling ($\rho = 0.6$–1.0) reaches it by about generation 21–23. The final sender–receiver gap also falls as coupling increases: from 0.033 without coupling to 0.010 at $\rho = 0.6$ and 0 at $\rho = 1.0$. Payoff peaks at $\rho = 0.6$, not at perfect coupling, suggesting that coupling helps align the code but complete lockstep is not necessarily the highest-payoff mutation structure in this finite-seed probe. The result supports the sender–receiver coordination hypothesis and implies that the harder problem is the joint transition in which comb architecture, orientation representation, and the sender–receiver code must all become useful together.

3.6 Long transition runs

We therefore ran longer transition experiments using the axial orientation representation. These runs used the baseline food distribution, 120 generations, ten seeds (101–110), sender–receiver transposition mutation correlation 0.6, three initial comb tilts (0, 0.5, and 1.0), and proportional vertical-comb advantages $\alpha \in \{0.05, 0.10, 0.15, 0.20, 0.25\}$. The default runs used $f(t) = t$, and a robustness check used $f(t) = \mathbf{1}[t \geq 0.8]$. Unlike the constrained near-vertical probe, comb tilt was free to mutate. A seed was counted as reaching the gravity channel when both mean sender and receiver transposition reached 0.50. It was counted as retaining verticality when final mean comb tilt was at least 0.80. It was counted as collapsed when mean foraging success fell to 0.02 or below at any generation, and as recovered when a collapsed seed ended with final success at least 0.10.

The proportional rule changes the interpretation of the long runs (Figure 1). From horizontal starts, no tested advantage reaches the gravity channel or retains verticality under either modifier, and final success remains high. From tilted starts, no tested advantage reaches the gravity channel or retains verticality either; a few seeds transiently collapse but recover by the final generation. Thus proportional verticality pressure up to 25% is not sufficient to bootstrap the architecture from horizontal or intermediate tilt under the current mutation and ecological settings.

From vertical starts, by contrast, verticality and gravity-referenced communication sometimes coevolve without lasting foraging failure. With $f(t) = t$, the number of seeds retaining verticality and reaching the gravity channel is 2/10 at $\alpha = 0.05$, 2/10 at $\alpha = 0.10$, 4/10 at $\alpha = 0.15$, 0/10 at $\alpha = 0.20$, and 3/10 at $\alpha = 0.25$. With the threshold modifier, the corresponding counts are 2/10 for $\alpha = 0.05$–0.20 and 3/10 at $\alpha = 0.25$. Collapse is limited to one seed in each vertical-start condition and those seeds recover by generation 120. The transition is therefore possible when vertical architecture is already supplied, but it remains stochastic and non-monotonic rather than robust.

3.7 Food-distribution transition probe

The long transition runs showed that the baseline food distribution does not bootstrap verticality from horizontal or tilted starts. We next asked whether more forgiving food distributions could create a viable transition corridor from horizontal starts without changing the inheritance or mutation rules. All runs in this probe started at $t = 0$, used axial orientation, retained the sender–receiver mutation correlation $\rho = 0.6$, used the proportional vertical-comb modifier $f(t) = t$ with $\alpha = 0.25$, and ran for 120 generations over five seeds (101–105). We varied only the food distribution: angular width, site count, capacity, and value.

The result is mixed but informative. Broader, richer, or higher-capacity patches substantially improve foraging success relative to the baseline, but they do not by themselves make the gravity code or stable vertical combs evolve from horizontal starts. For example, broad high-capacity patches raise final success to 0.391, but no seed reaches the gravity threshold or retains verticality, and mean final tilt remains 0.115.

The only successful horizontal-start transition in this probe occurs in the moderate-dense condition with four sites, angular width 0.30, capacity 10, and unit value. One of five seeds reaches

Cond.	Sites	Width	Cap.	Value	Stable	Grav.	Vert.	m_f	t_f	Succ.
Baseline	2	0.200	6	1.000	0/5	0/5	0/5	0.195	0.134	0.194
Broad	2	0.350	6	1.000	0/5	0/5	0/5	0.189	0.196	0.317
Broad cap.	2	0.350	12	1.000	0/5	0/5	0/5	0.230	0.115	0.391
Broad rich	2	0.350	6	1.500	0/5	0/5	0/5	0.195	0.197	0.316
Broad rich cap.	2	0.350	12	1.500	0/5	0/5	0/5	0.181	0.202	0.379
Dense	4	0.300	10	1.000	1/5	1/5	1/5	0.301	0.297	0.470

Table 6: Horizontal-start food-distribution transition probe. All rows use five seeds (101–105), axial orientation, 120 generations, sender–receiver mutation correlation $\rho = 0.6$, and proportional vertical-comb advantage $\alpha = 0.25$. Stable counts seeds that end with final comb tilt at least 0.80 and both sender and receiver transposition at least 0.50. Gravity counts seeds that reached the transposition threshold at any generation; vertical counts seeds retaining final comb tilt at least 0.80. Final m is the final mean of the lower of sender and receiver transposition.

the gravity-channel threshold at generation 75, reaches verticality at generation 108, and ends with comb tilt 0.911 and sender and receiver transposition 0.863 and 0.862. The same condition also has the highest final success (0.470). The other four seeds in that condition, however, stay far from vertical and do not reach the gravity threshold. Thus this ecology can open a transition path from horizontal starts, but the path is still rare rather than reliably selected under the current settings.

3.8 Local food-transition grid

To resolve the neighborhood around the seed-101 transition, we ran a local grid from horizontal starts while keeping the model rules fixed. The grid used seed 101, axial orientation, $\rho = 0.6$, food value 1.0, $f(t) = t$, 120 generations, and four worker processes. It varied the food-site count $n \in \{3, 4, 5\}$, angular width $w \in \{0.26, 0.30, 0.34\}$, capacity $K \in \{8, 10, 12\}$, and vertical-comb advantage $\alpha \in \{0.20, 0.25, 0.30\}$, for 81 conditions. Table 7 lists the only conditions that ended in a stable vertical gravity-code state.

Sites	Width	Cap.	α	Grav. gen.	Vert. gen.	m_f	t_f	Succ.
4	0.300	10	0.250	75	108	0.862	0.911	0.485
5	0.260	8	0.300	52	55	0.844	0.896	0.445
5	0.300	10	0.300	50	82	0.788	0.861	0.466
5	0.300	12	0.300	50	56	0.894	0.855	0.494

Table 7: Stable outcomes in the 81-condition seed-101 local ecology grid. All runs start horizontal, use axial orientation, $\rho = 0.6$, value 1.0, the proportional modifier $f(t) = t$, and 120 generations. Stable means final comb tilt at least 0.80 and both sender and receiver transposition at least 0.50.

The local grid confirms that the earlier transition is reproducible for the same seed and parameter setting, but it also shows that the transition is not a smooth monotone response to easier foraging or stronger verticality pressure. Only 4 of 81 seed-101 conditions are stable. The original pocket ($n = 4, w = 0.30, K = 10, \alpha = 0.25$) remains stable, while its immediate $\alpha = 0.20$ and $\alpha = 0.30$ neighbors do not. Three additional stable conditions occur with five food sites and $\alpha = 0.30$. In contrast, many high-success conditions at width 0.34 remain flat and do not reach the gravity threshold. The transition therefore appears to require a fairly specific balance: enough discovery and recruitment opportunity to keep foraging viable, but not so much ecological tolerance that vertical architecture and gravity-referenced transposition are weakly selected.

We then reran the four stable pockets over seeds 96–106 for 160 generations. This seed panel is a reproducibility check rather than a statistical neighborhood in the random-number sequence. Table 8 shows that the pockets remain rare.

Sites	Width	Cap.	α	Stable	Seeds	Mean m_f	Mean t_f	Mean succ.
4	0.300	10	0.250	1/11	101	0.238	0.244	0.457
5	0.260	8	0.300	2/11	100,101	0.300	0.369	0.424
5	0.300	10	0.300	1/11	101	0.264	0.229	0.512
5	0.300	12	0.300	1/11	101	0.258	0.245	0.506

Table 8: Robustness panel for the four stable pockets from the seed-101 local grid. Each pocket is rerun for seeds 96–106 and 160 generations. Stable counts final vertical gravity-code outcomes; the seed list gives which seeds were stable.

All four seed-101 stable states remain stable when extended to generation 160, so they are not transient generation-120 crossings. Across the 44 robustness runs, however, only five stable outcomes occur. The strongest pocket is $n = 5, w = 0.26, K = 8, \alpha = 0.30$, which succeeds in seeds 100 and 101. The other three pockets succeed only in seed 101. None of the robustness runs collapses, and final success remains high even in non-transition runs, so the limiting factor is not colony viability alone. Most failures keep foraging performance while either tilt stays low, transposition stays low, or both. This supports the view that the ecology can make the transition possible, but the architecture-and-code lock-in remains stochastic under the current mutation and selection regime.

3.9 Adaptive food-transition search

Manual grids become inefficient once several ecological parameters plausibly interact. We therefore added an Optuna-based adaptive search over the same scientific quantities, without introducing new biological mechanisms. The search starts from the long-transition configuration and keeps horizontal initial combs, axial orientation, $\rho = 0.6$, food value 1.0, $f(t) = t$, and 120 generations. It varies food-site count (3–6), angular width (0.22–0.36), capacity (6–14), vertical-comb advantage (0.15–0.35), maximum food distance (6.0–10.0), and travel cost (0.01–0.04). For each seed, let t_f be final comb tilt and let s_f, r_f be final sender and receiver transposition. A trial is stable when $t_f \geq 0.80$, $s_f \geq 0.50$, and $r_f \geq 0.50$. Its continuous near-miss score is

$$p = \min\left(1, \frac{t_f}{0.80}, \frac{s_f}{0.50}, \frac{r_f}{0.50}\right).$$

The optimization objective is stable seed count plus mean p, minus the number of seeds whose success falls to 0.02 or below. Thus an additional stable vertical gravity-code seed always dominates a merely better near miss, while the continuous term still lets the optimizer rank incomplete transitions.

As a pilot, we ran eight seed-101 trials with four workers. One trial reached the target state: six food sites, width 0.35, capacity 13, $\alpha = 0.33$, maximum food distance 8.0, and travel cost 0.01 ended with $t_f = 0.914$, $\min(s_f, r_f) = 0.855$, and final success 0.608. The other seven trials did not reach stability; their progress scores ranged from 0.129 to 0.319. This first adaptive pass therefore finds another viable seed-101 ecology, somewhat outside the earlier three-to-five-site local grid, but it should be treated as a search result rather than a robustness result until the best regions are rerun over multiple seeds.

We then continued the same Optuna study for 24 total trials over seeds 100–102, for 72 full seed simulations. No trial was stable in all three seeds, but the adaptive search did move beyond the single-seed result. The best trial was stable in two of three seeds, with six food sites, width 0.32, capacity 9, $\alpha = 0.34$, maximum food distance 6.0, and travel cost 0.03. It had objective 2.761, mean progress 0.761, mean final success 0.589, mean final comb tilt 0.669, and mean final minimum sender–receiver transposition 0.596. Three additional trials were stable in one of three seeds: the seed-101 pilot setting; a nearby six-site setting with width 0.32, capacity 9, $\alpha = 0.35$, maximum distance 6.0, and travel cost 0.03; and a lower-capacity six-site setting with width 0.28, capacity 7, $\alpha = 0.25$, maximum distance 6.0, and travel cost 0.025. None of the 24 trials collapsed. This suggests a more specific corridor than "easier food" alone: relatively many moderate patches, short food distances, and non-negligible travel costs can make vertical gravity-code transitions repeat across more than one seed, but the transition is still not reliable across the small seed panel.

3.10 Successful versus unsuccessful trajectories

To compare trajectories without confounding outcome with ecology, we reran the best Optuna pocket over a wider seed panel, seeds 96–120, and exported generation-level traces. The setting was six food sites, width 0.32, capacity 9, $\alpha = 0.34$, maximum food distance 6.0, travel cost 0.03, horizontal initial combs, axial orientation, $\rho = 0.6$, $f(t) = t$, and 120 generations. An exact rerun of one successful seed and one failed seed reproduced the saved event rows, confirming that identical seeds and identical configurations are deterministic in this implementation.

Five of 25 seeds ended in stable vertical gravity-code states: seeds 100, 102, 103, 108, and 114. The other 20 seeds did not collapse; their final foraging success was slightly higher on average than in the successful group (0.592 versus 0.585), so the dominant failure mode is not loss of viability. Instead, unsuccessful seeds mostly remain in a productive low-tilt, low-transposition basin. Successful seeds reached partial tilt ($t \geq 0.5$) by generation 39.0 on average, reached verticality by generation 65.4, and reached gravity-code transposition by generation 73.0. Their final mean tilt and minimum sender–receiver transposition were 0.884 and 0.829. Unsuccessful seeds had final means of only 0.278 and 0.207, although their final precision investment and receiver attention were higher than in the successful seeds.

The generation-level trajectories show the divergence. At generation 40, successful seeds already averaged tilt 0.553 and minimum transposition 0.250, whereas unsuccessful seeds averaged 0.332 and 0.186. By generation 80, the successful group had moved to tilt 0.760 and minimum transposition 0.641, while the unsuccessful group remained near tilt 0.288 and minimum transposition 0.183. Only one unsuccessful seed reached the gravity threshold at all; it did so late, at generation 104, crossed verticality at generation 116, and ended below the vertical threshold at $t_f = 0.721$. Thus the successful trajectory is not simply higher communication effort or higher foraging success. It is a coordinated path in which moderate verticality appears early enough for gravity-referenced transposition to become useful before selection settles into the flat direct-pointing regime.

4 Conclusion

The resource-distribution experiments support the idea that costly directional communication is favored most strongly in an intermediate ecological regime: food must be hard enough to make recruitment valuable, but discoverable enough for independent finders to seed dances. The geometry experiments support the architectural-precondition hypothesis in a more conditional way. A proportional vertical-comb advantage can sometimes preserve supplied verticality, and supplied near-vertical geometry allows gravity-referenced sender–receiver transposition to evolve, but the full architecture-and-code transition is not yet robust.

The constrained coupling probe supports the sender–receiver coordination hypothesis: with near-vertical geometry supplied, gravity-referenced communication evolves in every condition, and correlated sender–receiver mutations shorten the transition and reduce final mismatch. The long transition runs support the transition-risk hypothesis under a stricter viability-scaled fitness rule. Proportional vertical-comb advantages up to 25% do not bootstrap verticality from horizontal or tilted starts under the baseline ecology, and vertical starts produce gravity-referenced vertical outcomes only in a minority of seeds. The food-distribution probes suggest that ecology can matter: a moderate-dense environment produced one stable horizontal-to-vertical gravity-code transition in five seeds, and a local grid around that result found four seed-101 parameter pockets that sustain the transition. But simply making patches broader, richer, or higher-capacity mostly increases foraging success without making the architecture-and-code transition robust. The robustness panel reinforces this point: the best local pocket succeeds in only 2 of 11 seeds, and the other pockets succeed only in seed 101. The Optuna search found a stronger six-site short-distance pocket, but a wider seed panel still succeeds in only 5 of 25 seeds. Trajectory comparison shows that unsuccessful seeds usually retain high foraging success while remaining flat and direct-pointing. The next modeling priority is therefore to identify what converts rare viable paths into reliable selection: local refinement around the six-site short-distance corridor, altered mutation/coupling structure, or additional mechanisms that keep partially vertical combs available while gravity-referenced communication becomes useful.

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Modifier: $f(t) = t$

Gravity

	0.05	0.10	0.15	0.20	0.25
1.0	2/10	2/10	4/10	0/10	3/10
0.5	0/10	0/10	0/10	0/10	0/10
0.0	0/10	0/10	0/10	0/10	0/10

t_0 α

Vertical

	0.05	0.10	0.15	0.20	0.25
1.0	2/10	2/10	4/10	0/10	3/10
0.5	0/10	0/10	0/10	0/10	0/10
0.0	0/10	0/10	0/10	0/10	0/10

t_0 α

Collapse

	0.05	0.10	0.15	0.20	0.25
1.0	1/10	1/10	1/10	1/10	1/10
0.5	2/10	2/10	1/10	1/10	1/10
0.0	0/10	0/10	0/10	0/10	0/10

t_0 α

Recovered

	0.05	0.10	0.15	0.20	0.25
1.0	1/10	1/10	1/10	1/10	1/10
0.5	2/10	2/10	1/10	1/10	1/10
0.0	0/10	0/10	0/10	0/10	0/10

t_0 α

Final success

	0.05	0.10	0.15	0.20	0.25
1.0	0.19	0.20	0.20	0.20	0.20
0.5	0.20	0.20	0.20	0.20	0.20
0.0	0.20	0.20	0.20	0.20	0.20

t_0 α

Final tilt

	0.05	0.10	0.15	0.20	0.25
1.0	0.28	0.27	0.43	0.15	0.36
0.5	0.12	0.13	0.13	0.12	0.12
0.0	0.10	0.13	0.14	0.14	0.13

t_0 α

Modifier: $f(t) = \mathbf{1}[t \geq 0.8]$

Gravity

	0.05	0.10	0.15	0.20	0.25
1.0	2/10	2/10	2/10	2/10	3/10
0.5	0/10	0/10	0/10	0/10	0/10
0.0	0/10	0/10	0/10	0/10	0/10

t_0 α

Vertical

	0.05	0.10	0.15	0.20	0.25
1.0	2/10	2/10	2/10	2/10	3/10
0.5	0/10	0/10	0/10	0/10	0/10
0.0	0/10	0/10	0/10	0/10	0/10

t_0 α

Collapse

	0.05	0.10	0.15	0.20	0.25
1.0	1/10	1/10	1/10	1/10	1/10
0.5	1/10	1/10	1/10	1/10	1/10
0.0	0/10	0/10	0/10	0/10	0/10

t_0 α

Recovered

	0.05	0.10	0.15	0.20	0.25
1.0	1/10	1/10	1/10	1/10	1/10
0.5	1/10	1/10	1/10	1/10	1/10
0.0	0/10	0/10	0/10	0/10	0/10

t_0 α

Final success

	0.05	0.10	0.15	0.20	0.25
1.0	0.20	0.20	0.20	0.20	0.20
0.5	0.20	0.20	0.20	0.20	0.20
0.0	0.20	0.20	0.20	0.20	0.20

t_0 α

Final tilt

	0.05	0.10	0.15	0.20	0.25
1.0	0.27	0.28	0.27	0.27	0.36
0.5	0.11	0.11	0.11	0.11	0.11
0.0	0.12	0.12	0.12	0.12	0.12

t_0 α

Gravity: Seeds reaching gravity-channel transposition.
Vertical: Seeds retaining final comb tilt at least 0.80.
Collapse: Seeds whose success fell to 0.02 or below.
Recovered: Collapsed seeds ending with success at least 0.10.
Final success: Final-generation mean foraging success.
Final tilt: Final-generation mean comb tilt.

Figure 1: Long axial-orientation transition experiment. Rows are initial comb tilt, columns are proportional vertical-comb advantage α, and modifier headings give $f(t)$. Count panels report seeds out of ten; continuous panels report final-generation means. The generated fragment records the source CSV.